Introduction to Linear Algebra

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Agenda

- Introduction to system of linear equations
- Matrix Introduction
- Types of Matrix
- Basic Matrix Transformations
- Determinant
- Matrix Inverse
- Vector properties and operations
- Linear Independence
- Row Reduced Echelon Form

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What is Linear Algebra?

Linear Algebra is a branch of Mathematics that deals with symbols and rules that solve linear equations.

For example, Look at the following linear equations:

$$2a + 3b + 4c = 11$$

$$3a + 5b + 5c = 15$$

$$5a + 6b + 3c = 19$$

‘a’, ‘b’ and ‘c’ are the symbols and the equations are the rules.

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Linear Algebra Representation

Linear Algebra = Vectors & Matrices and operations on them.

Vectors and Matrices are objects in space with various dimensions.

The equations in the previous slide can be represented using matrix and vectors as:

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 5 & 6 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 11 \\ 15 \\ 19 \end{pmatrix}$$

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Application of Linear system

- Suppose an economy consists of the Coal, Electric (power), and Steel sectors, and the output of each sector is distributed among the various sectors as shown in Table 1, where the entries in a column represent the fractional parts of a sector's total output.
- The second column of Table 1, for instance, says that the total output of the Electric sector is divided as follows: 40% to Coal, 50% to Steel, and the remaining 10% to Electric. (Electric treats this 10% as an expense it incurs in order to operate its business.)
- Since all output must be taken into account, the decimal fractions in each column must sum to 1. Denote the prices (i.e., dollar values) of the total annual outputs of the Coal, Electric, and Steel sectors by $p_C, p_E, and\ p_S$, respectively. If possible, find equilibrium prices that make each sector's income match its expenditures.

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Application of Linear system

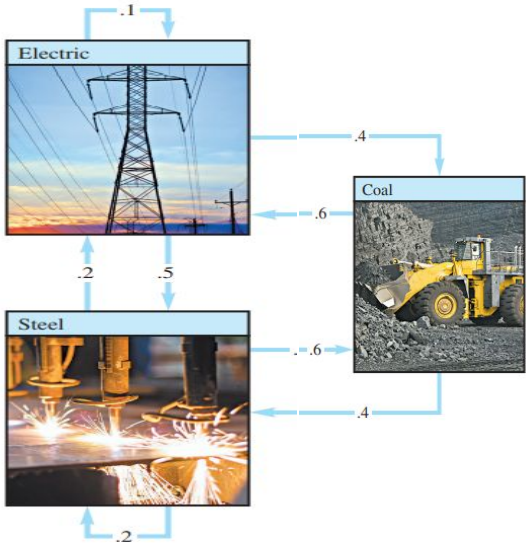

TABLE 1 A Simple Economy

Distribution of Output from:			
Coal	Electric	Steel	Purchased by:
.0	.4	.6	Coal
.6	.1	.2	Electric
.4	.5	.2	Steel

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Application of Linear system

- A sector looks down a column to see where its output goes, and it looks across a row to see what it needs as inputs. For instance, the first row of Table 1 says that Coal receives (and pays for) 40% of the Electric output and 60% of the Steel output.
- Since the respective values of the total outputs are p_E and p_S,
- Coal must spend $.4p_E$ dollars for its share of Electric's output and $.6p_S$ for its share of Steel's output.
- Thus Coal's total expenses are $.4p_E + .6p_S$.
- To make Coal's income, p_C, equal to its expenses, we want $pC = .4pE + .6pS$

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Application of Linear system

- $p_C = .4p_E + .6p_S$
- $p_E = .6p_C + .1p_E + .2p_S$
- $p_S = .4p_C + .5p_E + .2p_S$

$$
\begin{aligned}
p_C - .4p_E - .6p_S &= 0 \\
-.6p_C + .9p_E - .2p_S &= 0 \\
-.4p_C - .5p_E + .8p_S &= 0
\end{aligned}
$$

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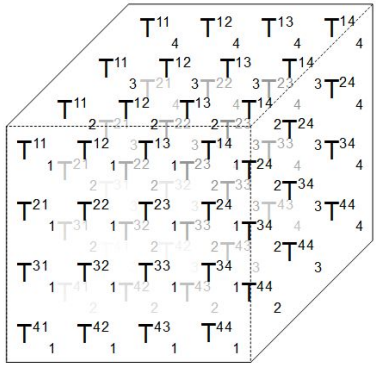
Basic Terms

- Scalar - Only Magnitude
- Vector - Direction & Magnitude - one dimensional array of numbers
- Matrices - Two dimensional array of numbers- can represent transforms/ multiple vectors
- Tensors- N Dimensional array of numbers

$$\left(T^{i\,j}{}_{k} \right)_{1 \le i,\, j,\, k \le 4} =$$



https://commons.wikimedia.org/wiki/File:Coordinates_of_a_tensor_of_type_(2,1)_in_dimension_4.png

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Example Scalar and Vectors

- 1.A box being pushed at a force of 125 newtons ANSWER: **scalar**
- 2. wind blowing at 20 knots ANSWER: **scalar**
- 3. a deer running 15 meters per second due west ANSWER: **vector**
- 4. a baseball thrown with a speed of 85 miles per hour ANSWER: **scalar**

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Introduction of Matrix

- A matrix is a 2 dimensional array of numbers arranged in rows and columns. Matrices provide a method of organizing, storing, and working with mathematical information.
- Matrices provide a useful tool for working with models based on systems of linear equations.
- If two matrices have the same size, they can be added or subtracted.
- If a matrix is multiplied by a scalar, each entry is multiplied by that scalar. We can consider scalar multiplication as multiplying a number and a matrix to obtain a new matrix as the product.
- To perform **multiplication of two matrices**, we should make sure that the number of columns in the 1st matrix is equal to the rows in the 2nd matrix.

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Introduction of Matrix (as Linear transformation)

- Matrix transform Space/ Vector
- Change of basis can be effected through a matrix transform

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 5 & 6 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 11 \\ 15 \\ 19 \end{pmatrix}$$

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Matrix Properties: Associative

$$A + B = B + A$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31}+b_{31} & a_{32}+b_{32} & a_{33}+b_{33} \end{pmatrix}$$

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Transpose

$$A \qquad\qquad\qquad\qquad A^T$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix}$$

If $A=A^T$, Then A is Symmetric Matrix

$$\begin{pmatrix} 1 & 3 & 4 \\ 3 & 2 & 5 \\ 4 & 5 & 8 \end{pmatrix}$$

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Identity Matrix

Identity matrix have no of rows same as no of columns

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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Other Types of Matrices

Lower Triangular Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 5 & 0 \\ 7 & 8 & 9 \end{bmatrix}$$

Upper Triangular Matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{bmatrix}$$

Diagonal Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Orthogonal Matrix

When the product of a square matrix and its transpose gives an identity matrix, then the square matrix is known as orthogonal matrix.

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

If $AA^T = A^TA = I$, then matrix A is orthogonal.

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Sparse Matrix

- Sparse matrix has its maximum entries zero

Dense Matrix

1	2	31	2	9	7	34	22	11	5
11	92	4	3	2	2	3	3	2	1
3	9	13	8	21	17	4	2	1	4
8	32	1	2	34	18	7	78	10	7
9	22	3	9	8	71	12	22	17	3
13	21	21	9	2	47	1	81	21	9
21	12	53	12	91	24	81	8	91	2
61	8	33	82	19	87	16	3	1	55
54	4	78	24	18	11	4	2	99	5
13	22	32	42	9	15	9	22	1	21

Sparse Matrix

1	.	3	.	9	.	3	.	.	.
11	.	4	.	.	.	.	.	2	1
.	.	1	.	.	.	4	.	1	.
8	.	.	.	3	1	.	.	.	.
.	.	.	9	.	.	1	.	17	.
13	21	.	9	2	47	1	81	21	9
.	.	.	.	.	.	.	.	.	.
.	.	.	.	19	8	16	.	.	55
54	4	.	.	.	11	.	.	.	.
.	.	2	.	.	.	.	22	.	21

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Applications of Matrices

User-rating matrix , it is sparse matrix why ?

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Basic Matrix Transformations

- Transformation matrix is a matrix that transforms one vector into another vector. The positional vector of a point is changed to another positional vector of a new point, with the help of a transformation matrix.

- Types:
  - Scale Transformation
  - Reflection Transformation
  - Projection Transformation
  - Rotation Transformation

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Scale Transformation

- A scaling transform changes the size of an object by expanding or contracting all vertices along the three axes by three scalar values specified in the matrix.

$$\mathbf{M} = \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- The s_x, s_y, and s_z values represent the scaling factor in the X, Y, and Z dimensions, respectively. Applying a scaling matrix to a point v produces an output vector with each component multiplied with the corresponding scaling value.

$$\begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \begin{pmatrix} s_x v_x \\ s_y v_y \\ s_z v_z \\ 1 \end{pmatrix}$$

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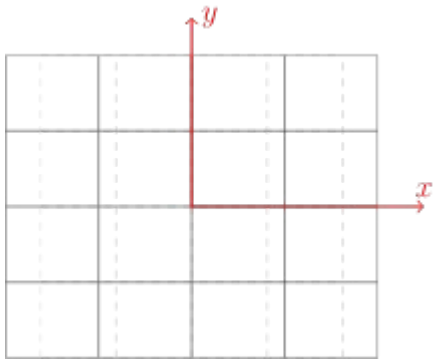
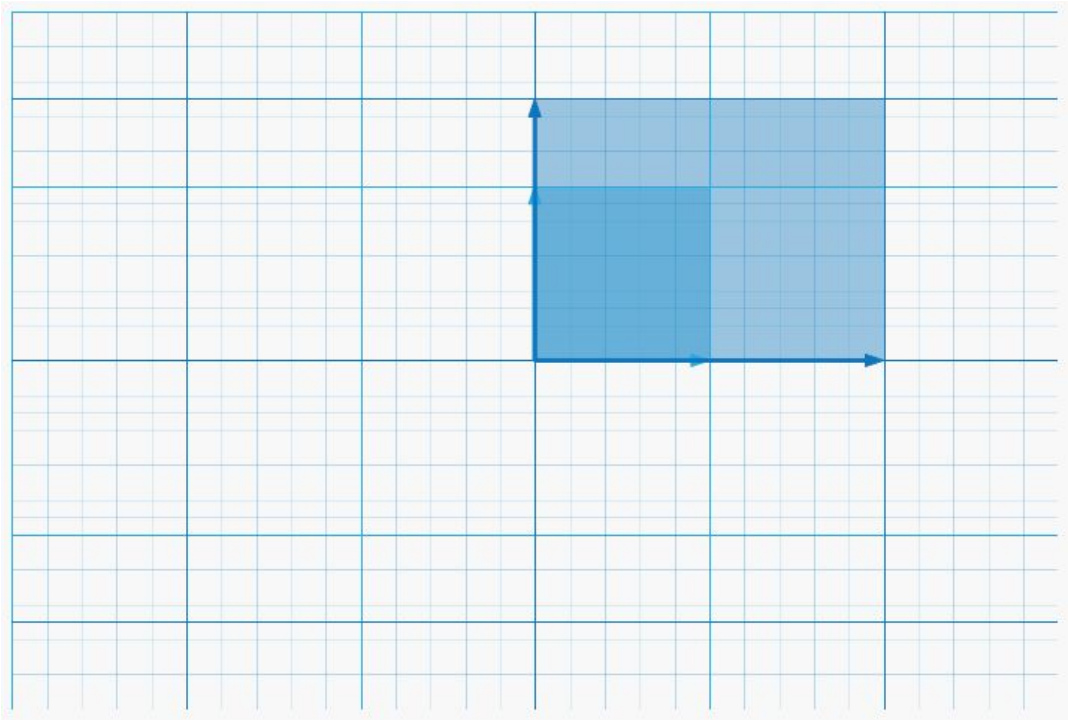
Scale Transformation





Transform

$$\begin{pmatrix} 2 & 0 \\ 0 & 1.5 \end{pmatrix}$$

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Reflection Transformation

- A reflection is a transformation that maps a figure to its reflection image. The figure on the right is the reflection image of a drawing and the point A over the line m. This transformation is called R_m, and we write A = R_m(A).

Reflection about y axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Reflection about x axis

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Reflection about the origin

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

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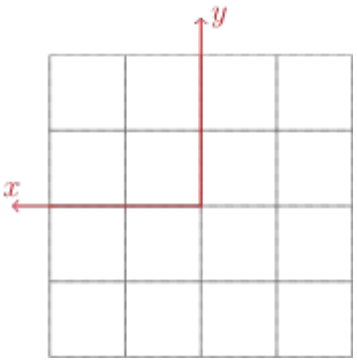
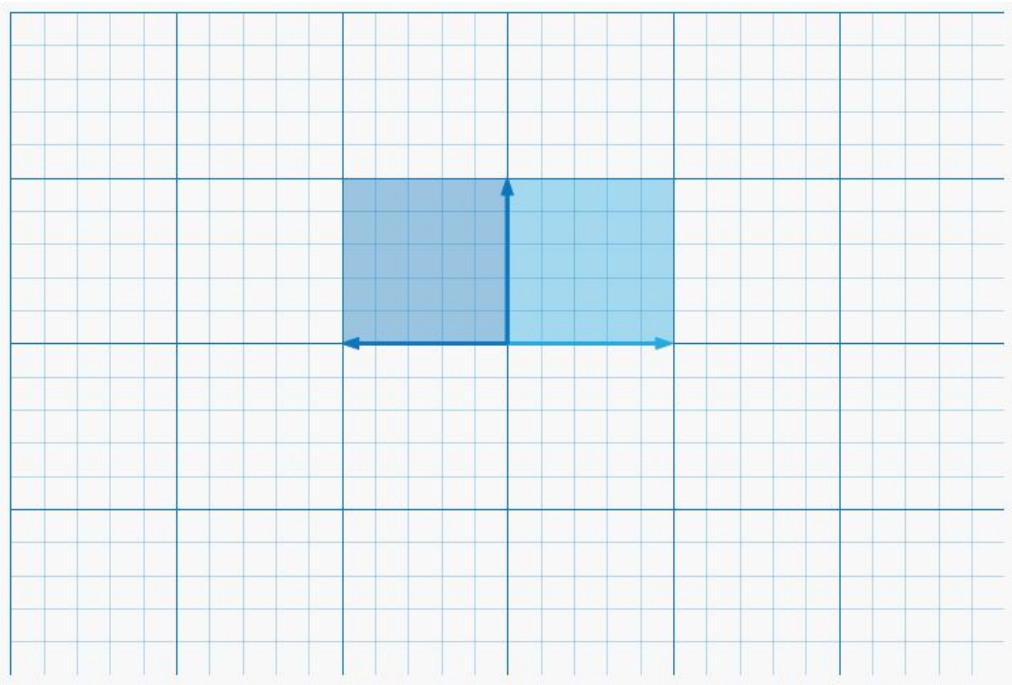
Reflection Transformation



Transform

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$



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Projection Transformation

- In linear algebra, a projection matrix is a matrix associated to a linear operator that maps vectors into their projections onto a subspace.

- The rule for this mapping is that every vector v is projected onto a vector T(v) on the line of the projection.

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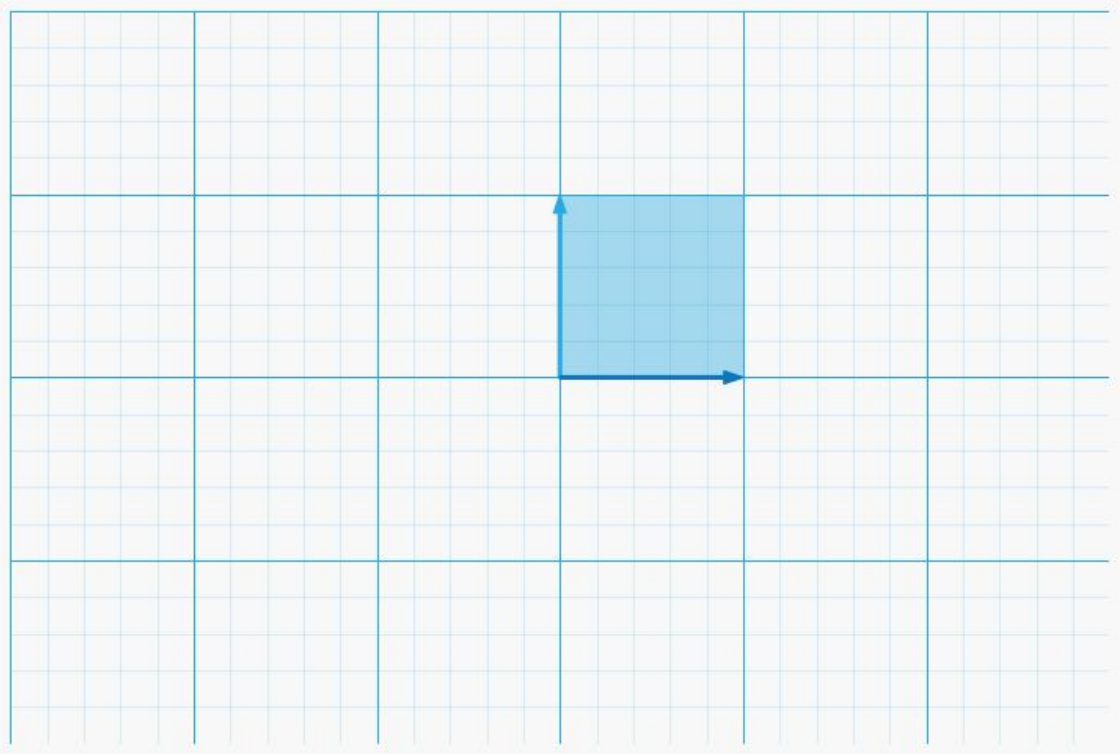
Projection Transformation



Transform

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

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Rotation Transformation

- A rotation matrix rotates an object about one of the three coordinate axes, or any arbitrary vector.
- The following three matrices $\mathbf{R}_X$, $\mathbf{R}_Y$ and $\mathbf{R}_Z$ and represent transformations that rotate points through the angle θ in radians about the coordinate origin.

$$\mathbf{R}_X(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{R}_Y(\theta) = \begin{pmatrix} \cos(\theta) & 0 & -\sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ \sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{R}_Z(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 & 0 \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

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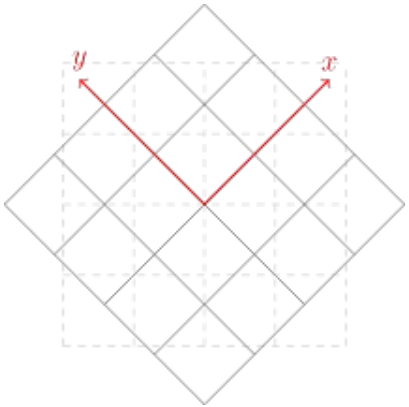
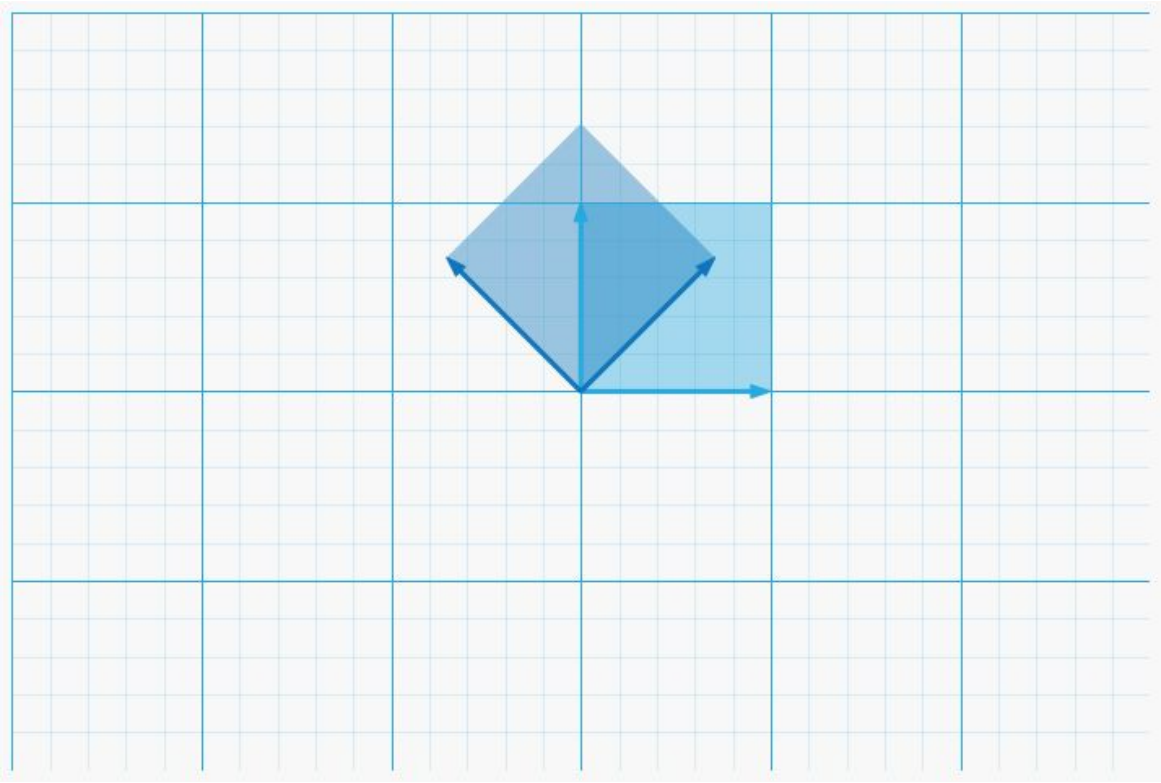
Rotation Transformation



Transform

$$\begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$



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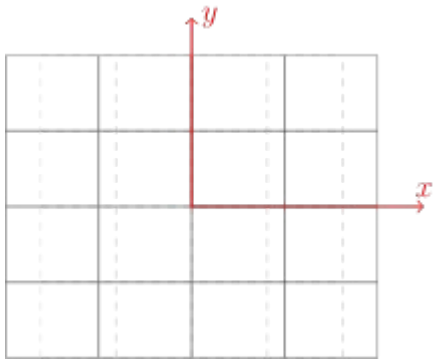
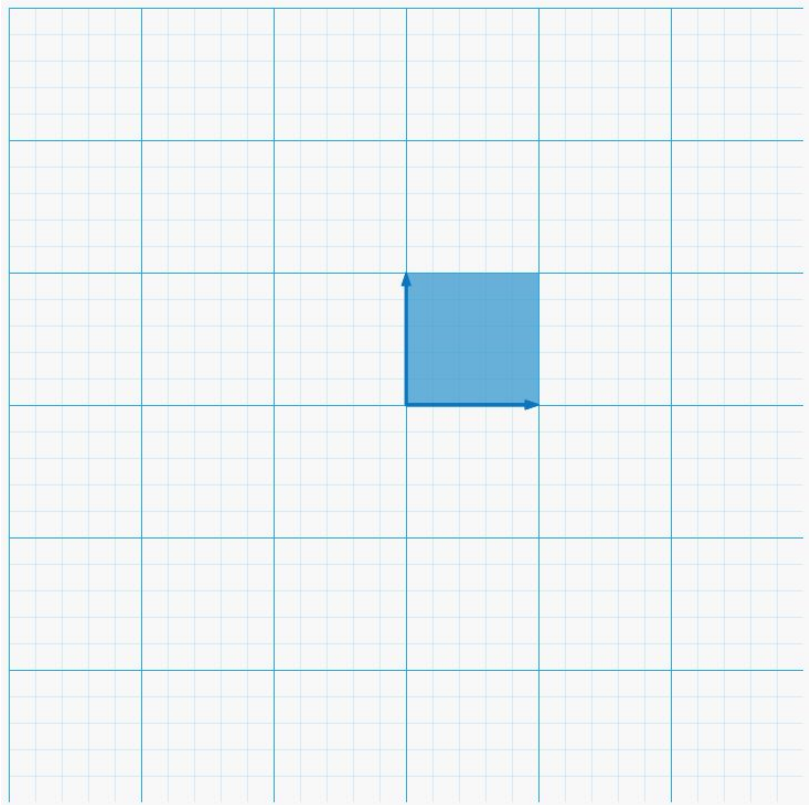
Identity Matrix Transformation





Transform

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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Identity Matrix & Matrix Inverse

As seen earlier, say for a 3 X 3 Matrix, Identity Matrix would be

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Matrix Inverse (A^{-1} is a Matrix which reverses/ nullifies the transformations caused by Matrix A

$$A\ I = A$$

$$A\ A^{-1} = I$$

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Physical Meaning of Matrix Inverse

$$\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\mathbf{A} \qquad \mathbf{X} \quad = \quad \mathbf{B}$$

- Applying a transformation of A on the vector X results in the vector B
- To find X, we need to apply a transformation on B which would inverse/reverse the effect of the transformation of A
- The $\mathbf{A}^{-1}$ transforms a inverse of the transformation introduced by A
- If we know the vector (B) in the transformed space, we can get the vector X by removing the effect of the transform

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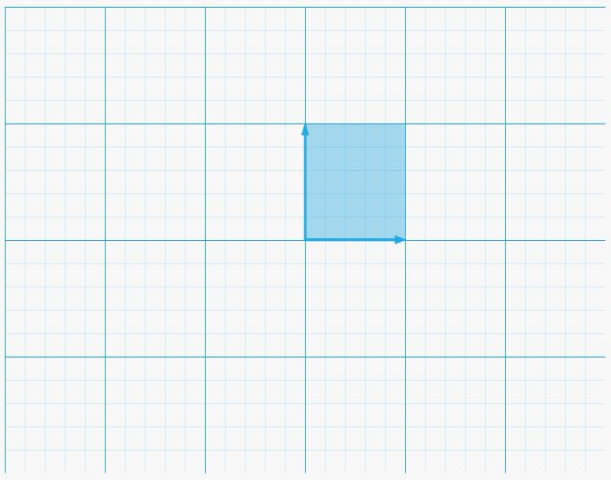
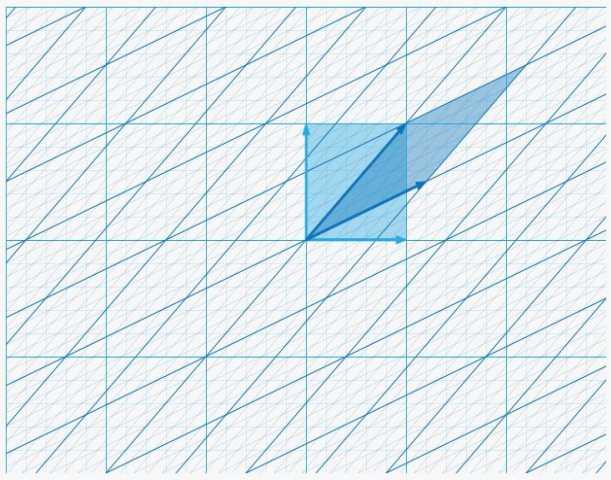
Physical Meaning of Matrix Inverse on Space

- Applying a transformation of A on the the space (Light blue) results in the space (Dark blue)





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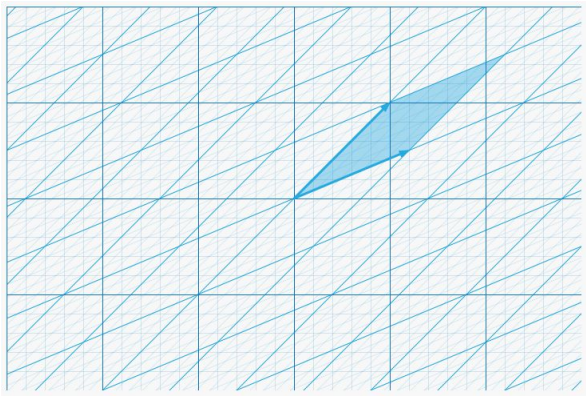
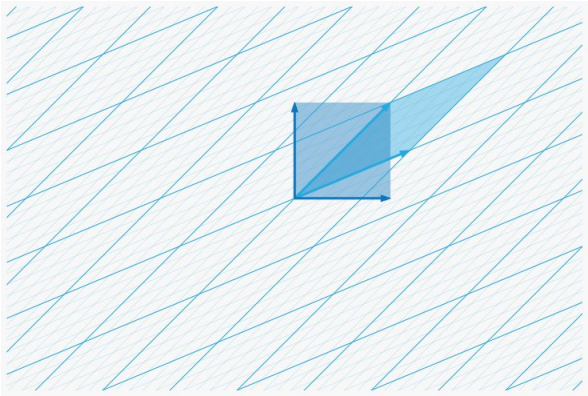
Physical Meaning of Matrix Inverse on Space





- Say, we have the transformed space, we need to apply a transformation on the same, which would inverse/reverse the effect of the transformation of A
- The $\mathbf{A}^{-1}$ transforms a inverse of the transformation introduced by A
- If we know the vector (B) in the transformed space, we can get the vector X by removing the effect of the transform

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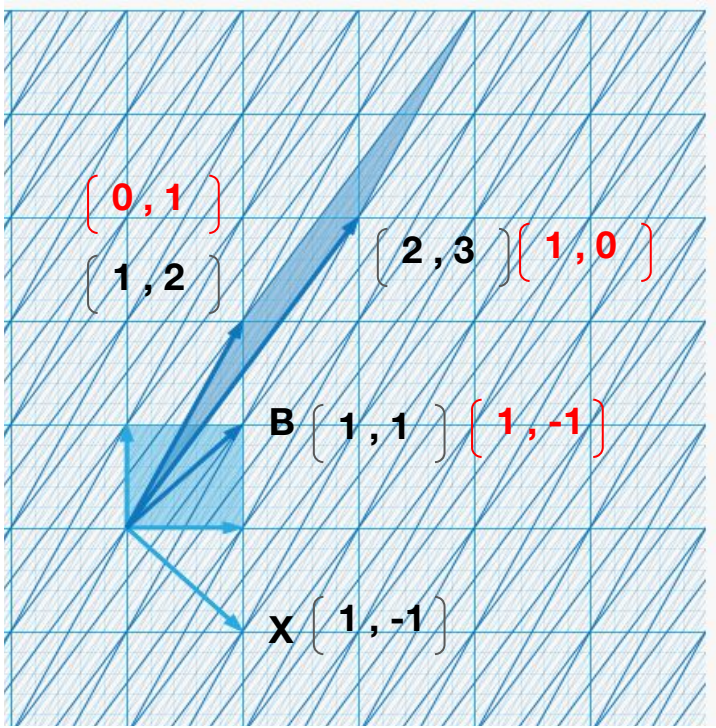
Physical Meaning of Matrix Inverse (Example)

$$\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \quad \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A \qquad X = B$$

$$A^{-1} \qquad B = X$$

$$\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$



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Determinant

Determinant for 2x2 matrix and 3x3 matrix - explanation with example

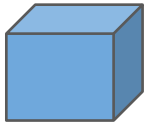
Intuition with area of graph transformed

Area when n = 2

$T:R^n \rightarrow R^n$. If you take a figure $S \subseteq R^n$, then $T(S) \subseteq R^n$.

For orientation preserving transformations :
1D - length,
2D - area
3D - volume

Volume when n=3



the content of T(S) will be the determinant times the content of S.
For an orientation reversing transformation the factor is the negation of the determinant.

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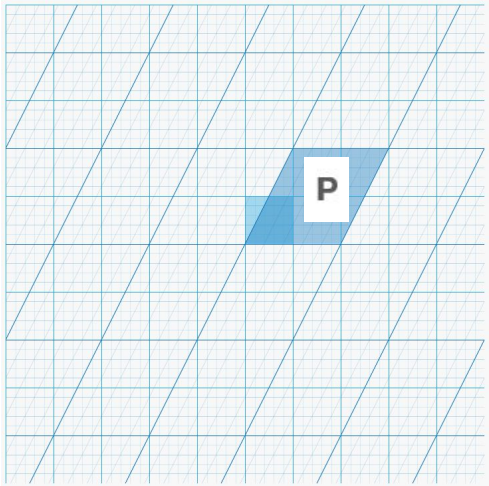
Determinant effect: Scaling of Area

- Consider The Following Transformation Matrix T applied on unit square which transforms unit square to parallelogram . Then The Area of parallelogram is Det(T)* area of unit square

$T = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ then Area(P) = Det(T) * area of unit square det(T) = 4



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Matrix Multiplication

$C_{m \times p} = A_{m \times n} B_{n \times p}$ $\quad c_{ij} = \Sigma a_{ik} b_{kj}$ for k=1 to n for each ij

A columns needs to be equal to B rows, to have a possible multiplication.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & & a_{mn} \end{pmatrix} \quad \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1p} \\ b_{21} & b_{22} & \dots & b_{2p} \\ \dots & \dots & \dots & \dots \\ b_{n1} & b_{n2} & & b_{np} \end{pmatrix}$$

$$c_{11} = a_{11} b_{11} + a_{12} b_{12} + \dots\dots \quad + a_{1n} b_{n1}$$

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Matrix Properties: Not Commutative

$$A_{n\times n} B_{n\times n} \neq B_{n\times n} A_{n\times n}$$

AxB need not be equal to BxA

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & & a_{nn} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & & b_{nn} \end{pmatrix} \neq \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & & b_{nn} \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & & a_{nn} \end{pmatrix}$$

$$c_{11}=a_{11}\, b_{11}+a_{12}\, b_{12}+\ldots\ldots\quad +a_{1n}\, b_{n1}$$

$$c_{11}=b_{11}\, a_{11}+b_{12}\, a_{12}+\ldots\ldots\quad +b_{1n}\, a_{n1}$$

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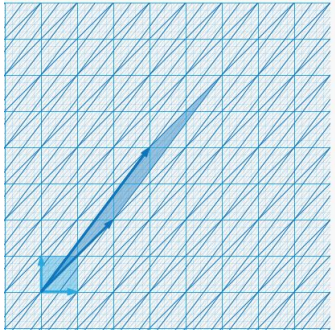
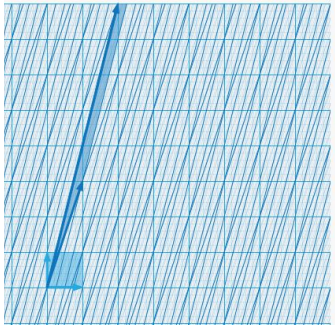
Matrix Properties: Not Commutative

$$A_{n\times n} B_{n\times n} \neq B_{n\times n} A_{n\times n}$$

$$\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \qquad \begin{pmatrix} 3 & 2 \\ 4 & 2 \end{pmatrix}$$



$$\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \qquad \begin{pmatrix} 2 & 1 \\ 8 & 3 \end{pmatrix}$$



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Matrix Operations (Dot/ Inner product)

$$\overset{A}{\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}} \quad \overset{B}{\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}} \quad \overset{C}{\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}}$$

A*(B*C)

$$\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 5 & 11 \end{pmatrix} = \begin{pmatrix} 7 & 17 \\ 8 & 20 \end{pmatrix}$$

(A*B)*C

$$\begin{pmatrix} 3 & 2 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 17 \\ 8 & 20 \end{pmatrix}$$

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Revisiting the Simultaneous Equations

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 5 & 6 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 11 \\ 15 \\ 19 \end{pmatrix}$$

$$A \qquad X = B$$

$$A\,X = B$$

$$A^{-1}(A\,X) = A^{-1}B$$

$$(A^{-1}A)X = A^{-1}B$$

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2D example for easy visualization

$$\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \qquad \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A \qquad X = B$$

$$A\,X = B$$

$$A^{-1}(A\,X) = A^{-1}B$$

$$(A^{-1}A)X = A^{-1}B$$

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Orthogonal Matrix

A matrix with orthonormal row and column vectors is called an orthogonal matrix.

Some useful properties:

- An orthogonal matrix Q is necessarily invertible (with inverse $Q^{-1} = Q^{T}$),
- The determinant of any orthogonal matrix is either +1 or −1.
- As a linear transformation, an orthogonal matrix preserves the inner product of vectors, such as a rotation, reflection or rotoreflection. In other words, it is a unitary transformation.

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Orthogonal Matrix (Examples)

$$\begin{array}{ccc} v_1 & v_2 & v_3 \\ \end{array}$$
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$v_1.v_2=v_2.v_3=v_1.v_3=0$

$|v_1|=|v_2|=|v_3|=0$

Det=1

$$\begin{array}{ccc} v_1 & v_2 & v_3 \\ \end{array}$$
$$\begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{pmatrix}$$

$v_1.v_2=v_2.v_3=v_1.v_3=0$

$|v_1|=|v_2|=|v_3|=0$

Det=-1

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Example of Vectors

- **Color**: (R,G,B)
- **Quantities of n different commodities** (or resources), e.g., bill of materials
- **Portfolio**: entries give shares (or $ value or fraction) held in each of n assets, with negative meaning short positions
- **Cash flow**: xi is payment in period i to us
- **Audio**: xi is the acoustic pressure at sample time i (sample times are spaced 1/44100 seconds apart)
- **Features**: xi is the value of ith feature or attribute of an entity
- **Customer purchase**: xi is the total $ purchase of product i by a customer over some period
- **Word count**: xi is the number of times word i appears in a document

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Word count vectors

- A short document: Word count vectors are used in computer based document analysis. Each entry of the word count vector is the number of times the associated dictionary word appears in the document.
- Consider The Following Examples
- Here We from set S of words excluding common words also called stop words like the etc etc S = {cat, dog, mat, sat, scratched} in dictionary order

(a)“The dog sat on the mat” is summarised by the vector w = (0 , 1 , 1 , 1 , 0).

(b) “The cat scratched the dog” is summarised by the vector w = (1, 1, 0, 0, 1).

(c) “The cat and dog sat on the mat” is summarised by the vector w = (1, 1, 1, 1, 0).

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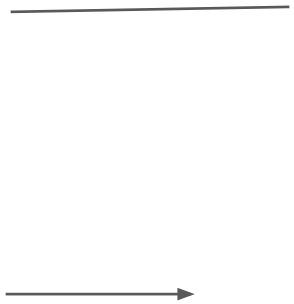
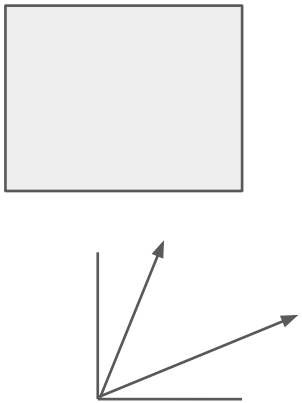
Vector Representations (Geometrically)

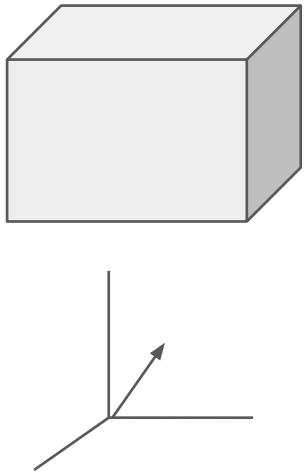
1 D



2 D



3 D



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Vectors (Data science Example)

Can be of very higher dimension (To the extent of number of features)

Iris Dataset:

Each flower has 4 features, which can be considered as 4 Dimensions and every row is a Vector

$$\begin{pmatrix} \text{Sepal_Length} \\ \text{Sepal_Width} \\ \text{Petal_Length} \\ \text{Petal Width} \end{pmatrix}$$

Flower	sepal_length	sepal_width	petal_length	petal_width
Flower1	5.1	3.5	1.4	0.2
Flower2	4.9	3.0	1.4	0.2
Flower3	4.7	3.2	1.3	0.2
Flower 1000	4.9	3.8	1.7	0.7

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Vector Representation

- Consider the following Data of Patient , Then measurement of Cholesterol , Systolic BP and Age in each row can be considered as a vector

Patient id	Cholesterol (mg/dl)	Systolic BP (mmHg)	Age (years)	Tail of the vector	Arrow-head of the vector
1	150	110	35	(0,0,0)	(150, 110, 35)
2	250	120	30	(0,0,0)	(250, 120, 30)
3	140	160	65	(0,0,0)	(140, 160, 65)
4	300	180	45	(0,0,0)	(300, 180, 45)

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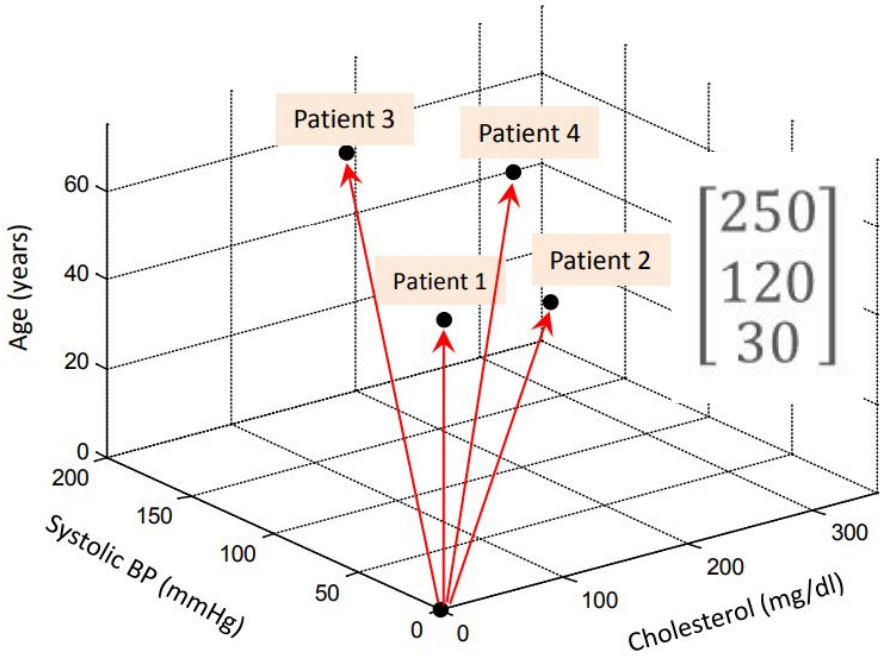
Vector Representation



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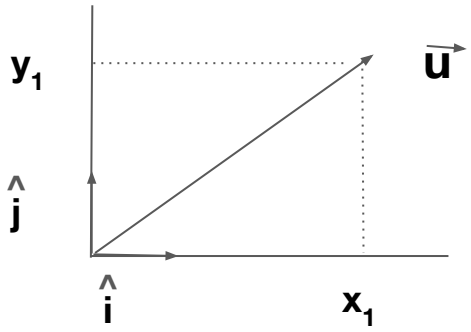
Coordinate System- Vectors

Symbol for vector : $\longrightarrow$ Symbol for unit vector : ^

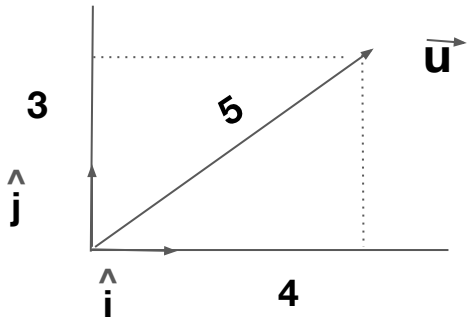


(x_1, y_1)

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$\|u\| = \sqrt{(x_1{}^2 + y_1{}^2}$$

$$\hat{u} = \frac{\vec{u}}{\|u\|} = \left(\frac{x_1}{\|u\|}, \frac{y_1}{\|u\|}\right)$$



$(4, 3)$

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\|u\| = \sqrt{(4^2 + 3^2}$$

$$\hat{u} = \frac{\vec{u}}{\|u\|} = \left(\frac{4}{5}, \frac{3}{5}\right)$$

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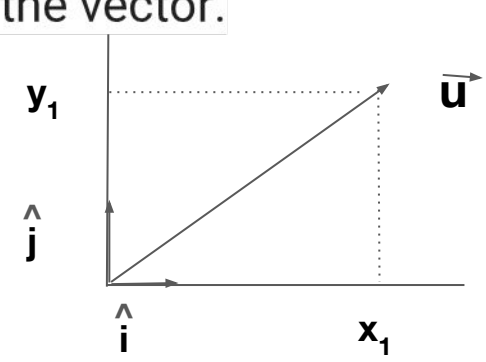
Magnitude of a vector - Vector norm

The length of vector is called the magnitude / norm of the vector.

There are Three norms - L1 and L2 and L_∞ Norms



$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

L1 Norm : $\|u\| = |x_1| + |y_1|$

L2 Norm : $\|u\| = \sqrt{(x_1{}^2 + y_1{}^2}$

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L_∞ Norm Chebyshev distance or Maximum Metric

- Given Vector $\mathrm{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$, then its L_∞ Norm is defined as follows
- $||u||_\infty = Max\{x_1, y_1\}$
- It can be also thought of as minimum number moves on chess boards for a king to move from one place to other

$\mathrm{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ then $||u||_\infty = Max\{2,3\} = 3$

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Minkowski distance

The Minkowski distance of order p (where p is an integer) between two points

$$X = (x_1, x_2, \ldots, x_n) \text{ and } Y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$$

is defined as:

$$D(X, Y) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}} \doteq \left|\left| \boldsymbol{x} - \boldsymbol{y} \right|\right|_{\boldsymbol{p}}$$

The General Norm is called **Minkowski distance** it is defined as for p =1 it becomes Manhattan for p= 2 it becomes Euclidean and for p = infinity it becomes Chebyshev distance

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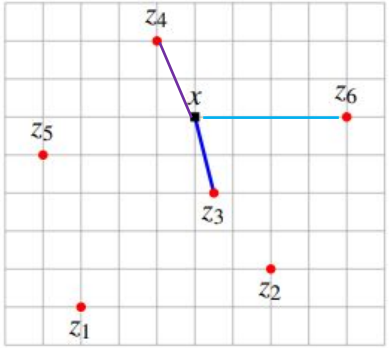
Application of distance: Nearest Neighbors

- Feature distance and nearest neighbors in the below diagram 3 nearest neighbor of x are z3 , z6, z4
  Based on Euclidean distances.

- if x and y are feature vectors for two entities, $\|x - y\|$ is the *feature distance*
- if $z_1, \ldots, z_m$ is a list of vectors, z_j is the *nearest neighbor* of x if

$$\|x - z_j\| \leq \|x - z_i\|, \quad i = 1, \ldots, m$$



- these simple ideas are very widely used

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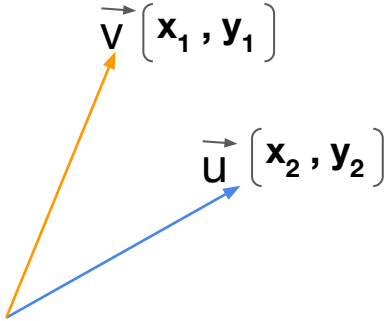
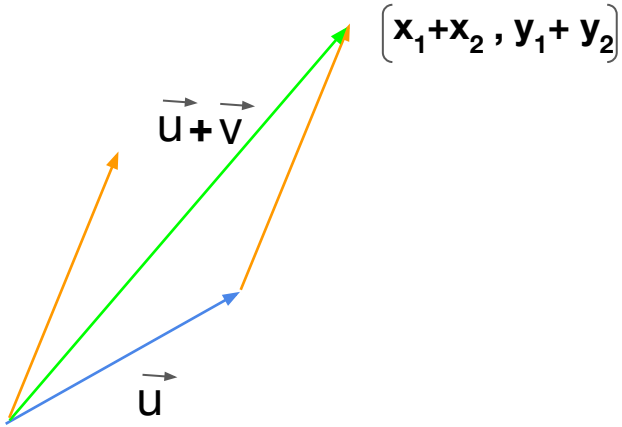
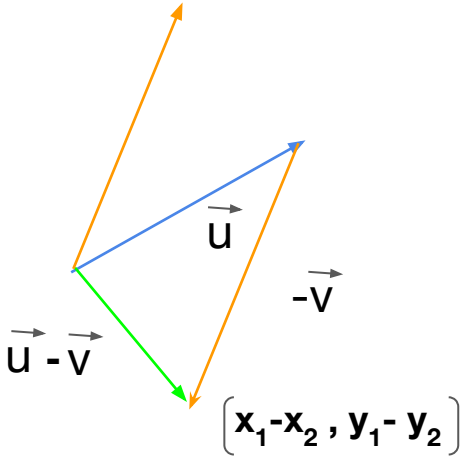
Vector Addition and subtraction







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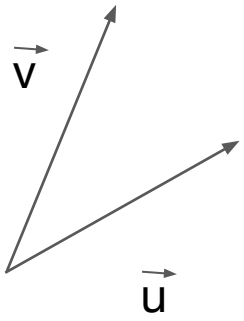
Vector Properties



- Associativity of addition $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w})$
- Commutativity of addition $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
- Distributivity of scalar multiplication with respect to vector addition $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$
- Multiplicative Identity is 1

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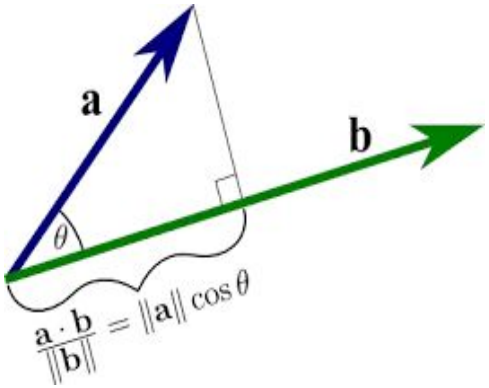
Vector Operations- Continued

Dot Product -

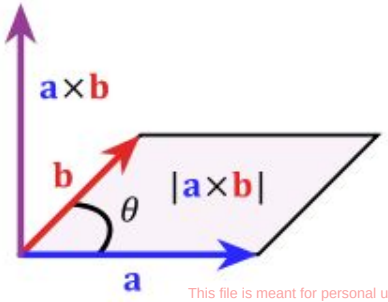
$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = a_1 b_1 + a_2 b_2$$

$$\begin{array}{c} \\ \vec{a_1} \rightarrow \\ \vec{a_2} \rightarrow \end{array} \begin{bmatrix} 1 & 7 \\ 2 & 4 \end{bmatrix} \cdot \begin{array}{c} \begin{array}{cc} \vec{b_1} & \vec{b_2} \\ \downarrow & \downarrow \end{array} \\ \begin{bmatrix} 3 & 3 \\ 5 & 2 \end{bmatrix} \end{array} = \begin{bmatrix} \vec{a_1} \cdot \vec{b_1} & \vec{a_1} \cdot \vec{b_2} \\ \vec{a_2} \cdot \vec{b_1} & \vec{a_2} \cdot \vec{b_2} \end{bmatrix}$$

$$A \qquad B \qquad C$$



Cross Product



The magnitude (length) of the cross product equals the area of a parallelogram with vectors **a** and **b** for sides.
Scalar transformations

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Dot Product Application

- Consider a firm operating two plants in two different locations. They both produce the same output (say, 10 units) using the same type of inputs. Although the amounts of inputs vary between the plants the output level is the same.
- The firm management suspects that the production cost in Plant 2 is higher than in Plant 1.
- The following information was collected from the managers of these plants.

PLANT 1

Input	Price	Amount used
Input 1	3	9
Input 2	5	10
Input 3	7	8

PLANT 2

Input	Price	Amount used
Input 1	4	8
Input 2	7	12
Input 3	3	9

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Dot Product Application

- Question 1. Does this information confirm the suspicion of the firm management?
- Plant 1: Price Vector $p_1 = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$ and Quantity Vector $q_1 = \begin{bmatrix} 9 \\ 10 \\ 8 \end{bmatrix}$
- Plant 2 : Price Vector $p_2 = \begin{bmatrix} 4 \\ 7 \\ 3 \end{bmatrix}$ and Quantity Vector $q_2 = \begin{bmatrix} 8 \\ 12 \\ 9 \end{bmatrix}$
- In Plant 1 the total cost is $\mathbf{c_1} = \mathbf{p_1}.\mathbf{q_1} = 133$, which implies that $\boldsymbol{unit\ cost\ is}\ \mathbf{13.3}$.
- In Plant 2, cost of production is $\boldsymbol{c_2} = \boldsymbol{p_2}.\boldsymbol{q_2} = 143$, which gives $\boldsymbol{unit\ cost\ as}\ \mathbf{14.3}$ which is higher than the first plant. That is, the suspicion is reasonable.

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Dot Product Application

- Question 2. The manager of the Plant 2 claims that the reason of the cost differences is the higher input prices in her region than in the other.
- Is the available information supports her claim?
- $p_2 = \lambda p_1$
- Plant I enjoys lower prices for inputs 2 and 3,
- whereas Plant 2 enjoys lower price for input 3.
- For a rough guess, one can still compare the lengths (norm)which are
- $||p_1|| = \sqrt{p_1 . p_1} = \sqrt{3^2 + 5^2 + 7^2} = \sqrt{83} = 9.11$ Similarly $||p_2|| = 8.60$

PLANT 1

Input	Price	Amount used
Input 1	3	9
Input 2	5	10
Input 3	7	8

PLANT 2

Input	Price	Amount used
Input 1	4	8
Input 2	7	12
Input 3	3	9

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Dot Product Application

- which indicates that price data does not support the claim of the manager of the Plant 2.
- When examined more closely, one can see that the Plant 2 uses the most expensive input (input 2) intensely.
- In contrast, Plant 2 managed to save from using the most expensive input (in this case input 3).
- Therefore, the manager needs to explain the reasons behind the choice mixture of inputs in her plant.

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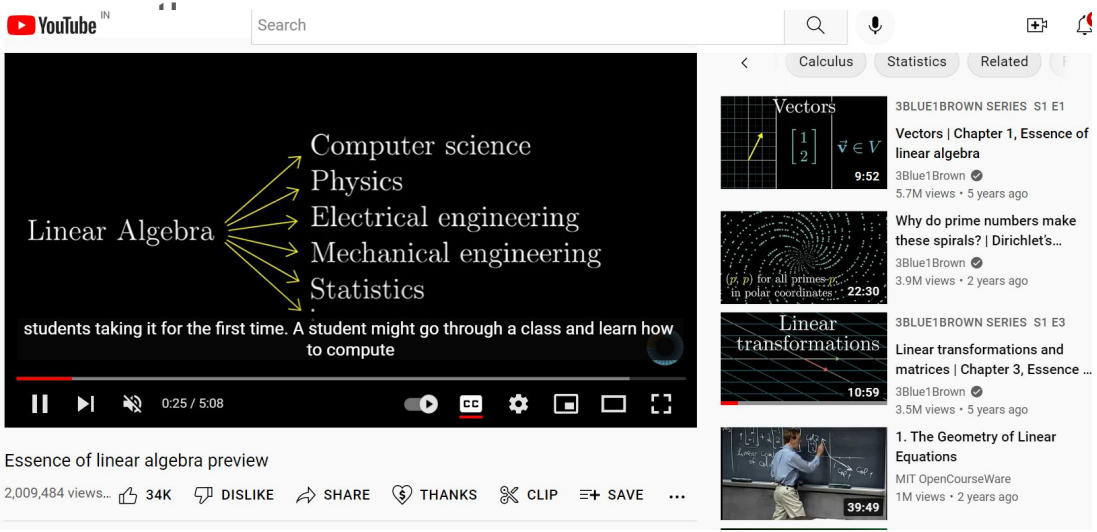
Dot Product Application : You tube video recommendation

Consider a user u and watching a video v on you tube , then user u can also see list of recommended videos (or very similar videos) in a window alongside. How does you tube recommend similar videos to the video user is watching



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Dot Product Application:

- Lets assume that we n users and m videos , in reality n, m can be in millions or billons. Consider the following arrangement of binary numbers o and 1 in which every row is a user and every column is a video , we define each entry as follows If user $u_i$watched video v_j then we fill that entry with 1 otherwise we fill it as 0 Let's assume we have 5 users and 9 videos

Ssimilarity between video vi and vj is calculated as vi.vj

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Dot Product Application

	v1	v2	v3	v4	v5	v6	v7	v8	v9
u1	1	1	0	1	0	1	0	0	0
u2	1	1	1	0	1	0	1	0	0
u3	1	0	0	0	1	0	0	0	1
u4	1	1	1	1	0	1	0	1	0
u5	1	0	0	1	0	1	0	1	0

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Dot Product Application

- Similarity between video v1 and v3 is , look at v1 column entries and v3 column entries and take a dot product

- $v1.v3 = [\,1\ 1\ 1\ 1\ 1\,].\begin{bmatrix}0\\1\\0\\1\\0\end{bmatrix} = 2$

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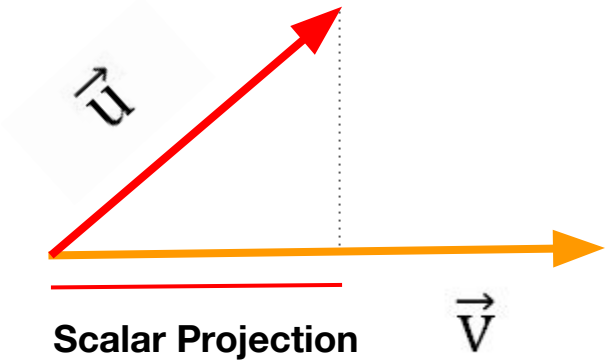
Scalar Projections

- A scalar projection of $\vec{u}$ onto $\vec{v}$ is the length of the shadow that $\vec{u}$ casts on $\vec{v}$:



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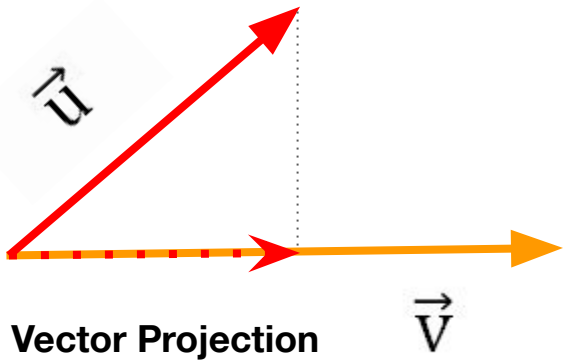
Vector Projections

- A vector projection of $\vec{\mathbf{u}}$ onto $\vec{\mathbf{v}}$ is the vector in the same direction as $\vec{\mathbf{v}}$, whose length is the scalar projection of $\vec{\mathbf{u}}$ on $\vec{\mathbf{v}}$.



- The scalar projection is the magnitude of the vector projection.

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Scalar and Vector Projections- Example

Find the scalar projection, the vector projection, and the orthogonal component of:

$(6\boldsymbol{i} + 7\boldsymbol{j})$ onto $(5\boldsymbol{i} - 12\boldsymbol{j})$

The scalar projection is:

$= 6\cdot 5 + 7\cdot(-12)$

$= -54/13$

The vector projection is:

$= -54(5i - 12j)/13^2$

$= -270i/169 + 648j/169$

The orthogonal component is:

$=(6i + 7j) -(-270i/169 + 648j/169)$

$=1284i/169- 535j/169$

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Linear Combination of vectors

- Consider the set of n vectors $S = \{v_1, v_2, \dots, v_n\}$ and scalars $c_1, c_2, \dots, c_n$ then

$c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$ is called linear combination of $v_1, v_2, \dots, v_n$

Eexamples :- suppose v1=$\begin{bmatrix}1\\2\end{bmatrix}$ and v2=$\begin{bmatrix}3\\4\end{bmatrix}$ c1 = 2, c2 = - 1.5

Then 2*v1 + (-1.5)*v2 = 2* $\begin{bmatrix}1\\2\end{bmatrix}$+(-1.5)* $\begin{bmatrix}3\\4\end{bmatrix}$ is linear combination of v1,v2

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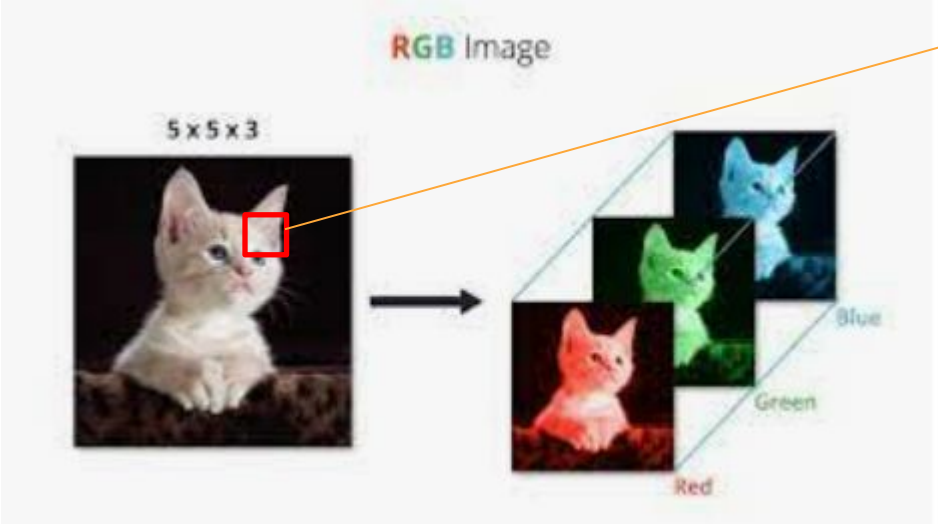
Example of linear combination

Consider color image also called RGB image , it has three component RED Green Blue As we can see red rectangle part of image is linear combination of RED GREEN and BLUE



C1*Red+c2*Green+c3*Blue

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Row Reduced Echelon Form (RREF)

A matrix is in RREF if:

1. Each nonzero row has a leading 1 (pivot).
2. Each pivot is the only nonzero entry in its column.
3. Pivots appear in a stair-step pattern (each to the right of the pivot in the row above).
4. Rows of all zeros appear at the bottom.

👉 Uniqueness: Every matrix has a unique RREF, regardless of row operation sequence.

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Linear Independence

- A Set of vectors is Linearly dependent, if one of the vectors can be defined as a combination of other vectors.
- If no vector of the set can be represented as combination of other vectors in the set, the set of vectors is linearly independent
- Hence, for vectors to be linearly independent, the linear combination of the vectors results in zero vector , if and only if all the constants are zero.

$$c_1\overrightarrow{v_1} + c_2\overrightarrow{v_2} + \cdots + c_n\overrightarrow{v_n} = 0 \qquad c_1 = c_2 = \cdots = c_n = 0$$

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Span of a Set

- Consider the set of n vectors $S = \{v_1, v_2, \dots, v_n\}$ and scalars $c_1, c_2, \dots, c_n$
- Then $L(S) = \{c_1 v_1 + c_2 v_2 + \cdots + c_n v_n : c_1, c_2, \dots, c_n$ scalar$\}$is called a span of S

Let S = $\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\}$ Then $L(S) = \{x\begin{bmatrix}1\\0\end{bmatrix} + y\begin{bmatrix}0\\1\end{bmatrix} : x, y\ scalars\}$

$L(S) = \begin{bmatrix}x\\y\end{bmatrix}$ for all real numbers x and y

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Basis Vectors

- If you can write every vector in a given space as a linear combination of some vectors and these vectors are independent of each other then we call them as basis vectors for that given space.
- **Properties of basis vector:**
- **Basis vectors must be linearly independent of each other.**
- **Basis vectors must span the whole space**
- **Basis vectors are not unique**
- **Basis vectors need not be unit vectors and orthogonal**

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Linear Independence (Example)

$$\begin{array}{c} R1 \\ R2 \\ R3 \end{array} \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 5 & 6 & 3 \end{pmatrix}$$

$$\begin{array}{c} R1 \\ R2 \\ R3 \end{array} \begin{pmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 5 & 6 & 3 \end{pmatrix}$$

R2= 2 R1
2R1-R2+0R3=0

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 5 & 7 & 9 \end{pmatrix} \begin{array}{c} R1 \\ R2 \\ R3 \end{array}$$

R1 + R2 = R3
1R1+1R2+(-1)R3=0

Only 2 vectors are sufficient, 3rd vector does not add any additional information

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Importance of Linear Independence

- Direction of two linearly dependent vectors is the same. This shows that there can be only as many solutions as the no. of independent vectors.
- Determines if the vectors form a basis
- Determines the no. of independent equations
- Determine the no. of independent vectors

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Basis

- Let S = $\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\}$ Then $L(S) = \{x\begin{bmatrix}1\\0\end{bmatrix} + y\begin{bmatrix}0\\1\end{bmatrix} : x, y\ scalars\}$

$L(S) = \begin{bmatrix}x\\y\end{bmatrix}$ for all real numbers x and y

Here aby vector in 2d plane can be written as linear combination of S.

S is Linearly independent as $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$ are not multiples of each other

Therefore S is a basis for 2d plane

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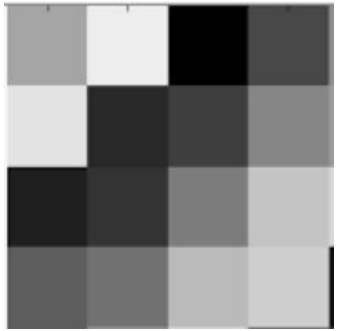
Basis Vector Application

- Consider The following Vectors v1 = $= \begin{bmatrix} 255 \\ 0 \\ 255 \\ 0 \end{bmatrix}$
- Vector v2 = $= \begin{bmatrix} 0 \\ 255 \\ 0 \\ 255 \end{bmatrix}$
- Then the image is linear combination of Vectors v1 and v2
- $= \frac{200}{255}$ $+ \frac{75}{255}$ $= \frac{200}{255} v_1 + \frac{75}{255} v_2$
- So v1 and v2 are basis vectors for any grey scale 4 by 4 image



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Appendix

- **numpy.linalg.norm(u)** # *Euclidean Distance; u is a array*
- **numpy.linalg.norm(u, ord=1)** # *Manhattan Distance; u is a array*
- ***Note:*** *linalg is a module in numpy to specifically work with linear algebra*

- Dot product: **numpy.dot(a, b)** *#Where a, b are two array/Matrix of same size*

- Scalar Projection: **sc_proj = np.dot(u, v) / np.linalg.norm(v)** *#Refer to the formula mentioned in the deck above*
- Vector Projection: **vec_proj = (v * sc_proj)/ np.linalg.norm(v)**

- **To inverse a matrix A:** V = inv(A)

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Thank You!

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